Project plan

Introduction

Background and Problem Statement

Current avionics equipment, even in the experimental and light-sport categories, remains prohibitively expensive for small operators, training fleets, and private owners. This high cost is driven primarily by low production volumes and the stringent hardware reliability requirements associated with certified life-critical systems.

There is a growing need for an affordable, internally developed alternative that can provide accurate position and orientation data without incurring the cost structure of certified or high-end experimental avionics. For Fumca RD, this need is operational rather than life-critical: the company requires a reliable logistics tracker to monitor where each aircraft is flying during daily spraying runs, without the $20,000+ price tag of certified units such as the Satloc Falcon Pro.

Company Background

Fumca RD is a Dominican-based agricultural aviation company specializing in advanced crop-dusting technology. It operates in the northwest of the island, a region where the local economy is dominated by agriculture. The company is widely recognized for the precision and reliability of its aircraft, flight crews, and operations, and it is the largest general aviation (GA) fuel consumer in the Caribbean.
Over the last two decades, the Engineering department has rebuilt multiple Cessna 188s from donor airframes. These aircraft have been optimized for increased flight performance and fitted with leading-edge avionics that use sensor fusion and GPS imaging data to achieve optimal spray coverage for customer requirements.

Project Context

This graduation assignment investigates the feasibility of developing a low-cost embedded navigation unit by fusing inertial measurement data with GPS. The system targets sub-3-meter position accuracy and reliable real-time attitude estimation using commercial off-the-shelf sensors and microcontrollers. The project will be executed at Fumca RD's Engineering & Technical Department in Piloto, Dominican Republic, over a 20-week period.

Organization Structure

Fumca RD is divided into several departments. The core business is agricultural spraying, supported by flight operations (pilots and logistics coordinators who manage schedules and aircraft availability) and an Engineering & Technical Department responsible for airframe construction, maintenance, and avionics integration. The Engineering & Technical Department is the primary host for this graduation project. The department comprises engineers, technicians, and certification contractors. Within this department, my work will focus on avionics and aircraft systems. I will be embedded with the company's avionics integration specialist and will consult Satloc's avionics engineers for advanced embedded development guidance.

My company supervisor is Gaby Matias, one of the company's co-founders. He is a highly experienced pilot with decades of combined operational and technical expertise in agricultural aviation. Under his supervision, I will be provided with a selection of sensor kits for evaluation, comparison, and integration into the aircraft. I will collaborate closely with the in-house avionics integration specialist for hardware installation and airworthiness considerations, and with Satloc engineers for firmware architecture and embedded systems support.

Project goal & objectives

Project goal

The goal of this project is to investigate the feasibility of developing a low cost embedded intertial navigationn system / golbal navigation satelite system (INS/GNSS) fusion unit for fumca RD experimental cessna 188 fleet. The system shall use commertial off shelf inetrial measurement units, GNSS receivers and bare metal microcontroller firmware to provide real-time position and attitude data. The aim is to determine whether a non certified, internally developed soplution can achieve sufficient accuracy for logistics tracking and operational data collection at a fraction of the cost of certified avionics.

Project objectives:

ID	Objective	SMART Justification
O1	**Develop and simulate a sensor-fusion algorithm** capable of fusing IMU and GNSS data into a unified position, velocity, and attitude solution, using quaternion-based rotation to avoid gimbal lock.	*Specific:* Algorithm type defined. *Measurable:* Simulation outputs vs. reference trajectory. *Achievable:* Based on existing C & embedded skills. *Relevant:* Core technical requirement. *Time-bound:* Completed by Week ().
O2	**Implement the fusion algorithm in embedded C** on a bare-metal MCU (RP2350, STM32, or NXP LPC) using vendor SDKs and the arm-none-eabi-gcc toolchain, with no Arduino abstraction layers.	*Specific:* Language, toolchain, and MCU family defined. *Measurable:* Firmware compiles and executes. *Achievable:* Prior embedded systems experience. *Relevant:* Smart systems competency. *Time-bound:* Completed by Week ().
O3	**Design and integrate a prototype hardware assembly** including the selected IMU(s) (LSM6DS3, Bosch BMI-series, or ADIS/ADXL) and GNSS module (NEO-M9N), with physical mounting and cabling suitable for installation in a Cessna 188.	*Specific:* Hardware components listed. *Measurable:* Physical prototype installed and powered. *Achievable:* Supported by Fumca RD avionics specialist. *Relevant:* Deliverable for company. *Time-bound:* Completed by Week ().
O4	**Validate position accuracy to within 3 meters (horizontal, 95 % confidence)** under operational crop-dusting conditions by comparing prototype output against the Garmin G500 and Satloc Falcon Pro during at least three flight sessions on the Cessna 188 or T303.	*Specific:* 3 m threshold and reference units defined. *Measurable:* Statistical comparison of logged data sets. *Achievable:* Flight access confirmed by company. *Relevant:* Directly answers main research question. *Time-bound:* Completed by Week ().
O5	**Quantify the cost-to-accuracy trade-off** of the COTS solution relative to certified avionics, including bill-of-materials, development time, and accuracy degradation under high-G and high-bank-angle maneuvers typical of crop-dusting.	*Specific:* Cost and accuracy metrics defined. *Measurable:* Cost table and error budget analysis. *Achievable:* Company provides Satloc cost data. *Relevant:* Core research question. *Time-bound:* Completed by Week ().
O6	**Document all work in a graduation thesis** that includes theoretical framework, methodology, results, and recommendations for future Fumca RD development, and deliver a complete, version-controlled project archive (firmware source, schematics, test data).	*Specific:* Deliverable format defined. *Measurable:* Submission to Onstage. *Achievable:* Standard graduation requirement. *Relevant:* Academic assessment. *Time-bound:* Completed by Week ().

Deliverables

At the end of the 20-week project, the following tangible deliverables will be handed over:

1. **Graduation Thesis** A complete technical report documenting the literature review, theoretical framework, design decisions, test results, and cost-to-accuracy analysis.

2. **Functional Embedded Prototype** A physical sensor-fusion unit (PCB, sensors, enclosure if applicable) capable of logging real-time fused navigation data.

3. **Firmware Repository** Complete source code in C, built with CMake and arm-none-eabi-gcc, including drivers for the selected IMU and GNSS, the fusion filter implementation, and data-logging routines. Version-controlled via Git.

4. **Validation Data Set** Synchronized logs from the prototype, Garmin G500, and Satloc Falcon Pro, collected during flight tests, plus a comparison report.

5. **Project Archive** All schematics, datasheets, simulation scripts (Python/MATLAB/Desmos), and the final project plan with execution evidence.

Scope & Boundaries

In scope:

- Algorithm design, simulation, and embedded implementation.
- Ground testing (vibration, EMI, static accuracy) at Fumca RD's Piloto facility.
- In-flight validation on non-revenue flights (Cessna 188 or T303).
- Cost analysis and comparison with certified avionics.
- Bare-metal firmware development (no Arduino framework).
- Airframe modifications for Avionics implementation.

Out of scope:

- Certification to any aviation authority standard (FAA, EASA, IDAC/ICAO). The device is explicitly non-life-critical and experimental.
- Cloud infrastructure, real-time fleet tracking dashboard, or pilot user interface. The project stops at the embedded data-logger unit.
- Development of custom ASICs or non-COTS sensors.

Main & sub research questions

Main:

- What is the cost-to-accuracy trade-off of implementing a INS/GPS fusion system on commercial off-the-shelf embedded hardware, relative to certified avionics solutions?

Sub:

- Where do main cost go into modern commercial avionics such as the Satloc Falcon pro?
- How does loosely coupled filters on off-shelf sensors and controllers compare to direct GPS readings on certified equipment?
- How does (pitch, roll, yaw) uncertainty effect the positioning uncertainty over time?
- Other than positional accuracy in what other conditions does having IMU + GPS benefit the purpose of the device?
- What existing solutions exist (at consumer grade) and what should be considered when this method is applied on a Cessna 188 or Air Tractor?
- How does environment compensation such as vibration and EMI insulation effect the total error budget?
- What are the single point failure modes in INS/GPS fusion system that certified equipment negate?
- What other sensors (such as time of flight) can be used for a better prediction and what other issues may arise from their implementation?
- What existing data can be used to validate the results from the device?

Answering research Quesrtions (planning)

Sub-RQ	Method	Instrument / Tool	Deliverable
RQ1: Cost breakdown of certified avionics	Literature review & market analysis	Satloc/Garmin datasheets, interviews with Fumca RD procurement	Cost comparison table
RQ2: Loosely coupled vs. certified accuracy	Experimental comparison	COTS prototype + Garmin G500 + Satloc Falcon Pro	Accuracy comparison report with statistical analysis
RQ3: Orientation uncertainty propagation	Simulation & theoretical analysis	MATLAB/Python error-state model	Error propagation graph / covariance analysis
RQ4: Benefits beyond position	Literature & experimental	Flight logs showing GPS dropout recovery	Use-case analysis report
RQ5: Consumer-grade solutions for Cessna 188	Literature & benchmarking	ArduPilot, PX4, open-source INS papers	Feasibility assessment report
RQ6: Vibration/EMI compensation effect	Experimental	Oscilloscope, logic analyzer, vibration tests on T303	Error budget before/after mitigation
RQ7: Single-point failure modes	FMEA / theoretical analysis	System architecture diagrams	Risk register / failure mode table
RQ8: Auxiliary sensors (ToF, etc.)	Literature & prototyping	Sensor datasheets, brief integration test	Recommendation report with trade-offs
RQ9: Validation data sources	Comparative analysis	Garmin G500 logs, Satloc logs, Cessna 188 flight data	Validation methodology document

To answer the main research question, a mixed-methods approach is used. The cost analysis (RQ1) is addressed through literature and procurement data. The technical performance questions (RQ2–RQ4, RQ6) are answered through simulation and in-flight experimentation. System-level questions (RQ5, RQ7–RQ8) are addressed through literature review and failure-mode analysis. Validation (RQ9) is ensured by direct comparison against certified reference equipment during flight testing.

Planning

Phase Breakdown

Phase 1: Initiation & Literature
This phase consists of establishing the theoretical and commercial foundation for the geolocalizer. Activities include reviewing literature on INS/GPS loosely-coupled fusion, quaternion-based attitude estimation, and error-state Kalman filtering. I will survey consumer-grade IMUs (LSM6DS3, Bosch BMI-series, ADIS/ADXL) and GNSS modules (NEO-M9N) to determine viability for high-G, high-vibration crop-dusting environments. The phase also includes a cost-breakdown analysis of certified avionics (Satloc Falcon Pro, Garmin G500) to define the target accuracy-to-cost trade-off.

Phase 2: Design & Simulation
This phase is the architectural backbone of the project. Based on the selected sensors and MCU (RP2350, STM32, or NXP LPC), I will define the data pipeline: IMU sampling frequency, GNSS update rate, timestamp synchronization strategy (burst vs. stream), and communication interfaces. The sensor-fusion algorithm (Madgwick, Mahony, or extended Kalman) will be prototyped in simulation using realistic process noise derived from Cessna 188 flight dynamics (high bank angles, low-altitude GPS multipath). Any architectural errors identified here must be resolved before hardware commitment to avoid costly redesign in later phases.

Phase 3: Hardware & Firmware
This phase moves from simulation to physical implementation. It consists of two parallel tracks: (1) hardware design or adapting a sensor carrier board for installation in the Cessna 188, including vibration isolation and EMI shielding and (2) firmware development bare-metal embedded software using arm-none-eabi-gcc, CMake, and vendor SDKs (no Arduino abstraction). The firmware must initialize the IMU and GNSS, execute the selected fusion filter in real time, and log raw and fused data to non-volatile storage for post-flight analysis. OpenOCD and a logic analyzer/oscilloscope will be used for debugging.

Phase 4: Testing & Validation
This is the longest phase because it consists of iterative ground testing followed by in-flight validation. Ground testing includes static noise characterization, vibration-table or engine-run-up tests on the Cessna T303 (same engines as the 188), and EMI immunity checks at the Piloto hangar. Flight testing requires coordination with Fumca RD logistics to secure non-revenue aircraft slots. The prototype will be installed alongside the Garmin G500, and data will be batch-logged for direct comparison against the Garmin and Satloc Falcon Pro outputs. Special attention will be paid to GPS dropout scenarios during low-altitude turns and high-G pull-ups.

Phase 5: Analysis & Reporting
This phase consists of analyzing the collected data to answer the main and sub-research questions. I will quantify the cost-to-accuracy trade-off of the COTS solution relative to certified avionics, assess how orientation uncertainty propagates into position error over time, and evaluate the effectiveness of vibration/EMI compensation. The results will be documented in the graduation thesis, including the theoretical framework, methodology, results, and recommendations for Fumca RD. All code, schematics, and raw data will be archived in a version-controlled repository.

Phase 6: Buffer / Finalization
This phase is reserved as contingency time for delays accumulated in earlier phases (e.g., flight-test weather cancellations, sensor delivery issues, or filter tuning iterations). It also includes integrating feedback from the university coach, finalizing the thesis, preparing the final presentation, and delivering the completed project plan execution evidence for assessment. If the buffer is unused, the time will be allocated to extended flight-test repetition for higher statistical confidence.

Use case for the company

Problem description
Research topics
Project goal/objective
Main & sub research questions
Planning (Gantt chart)
Relation between competencies and assignment